Lecture 2 — Multiple Regression

Introductory Econometrics — Télécom Paris

Màxim Sandiumenge i Boy

Reading: Wooldridge (5th ed.) ch. 3

Roadmap

- Lecture 1 used simple regression to relate y to one explanatory variable x.
- A ceteris paribus interpretation requires other factors affecting y to be unrelated to x.
- Multiple regression lets us control explicitly for several factors that affect y simultaneously.

1. The multiple linear regression model
2. Ordinary least squares in scalar and matrix form
3. Partialling out and goodness of fit
4. Unbiasedness, variance, and Gauss–Markov
5. Multicollinearity and interpretation

Motivating example

Suppose we want the effect of education on wages while holding labour-market experience fixed:

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u. \tag{3.1}$$

- Compared with a simple regression, equation (3.1) takes experience out of the error term.
- Then we do not need to assume that education and experience are uncorrelated.
- β_1 compares workers with different education but the same included controls.
- u contains ability, family background, motivation, and measurement errors.
- A causal interpretation still requires a credible argument that education is unrelated to the remaining unobservables after conditioning on the controls.

The multiple linear regression model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u. \tag{3.6}$$

- The model is linear in parameters; regressors can be transformed, dummies, or interactions.
- The error u collects other determinants of y.
- The key assumption is
$$\mathbb{E}(u \mid x_1, \ldots, x_k) = 0. \tag{3.8}$$
- Hence the population regression function is
$$\mathbb{E}(y \mid x_1, \ldots, x_k) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k.$$
- A causal effect requires correct specification and a credible argument that remaining unobservables do not confound the comparison.

Why simple regression can fail

Suppose the correctly specified population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \qquad \mathbb{E}(u \mid x_1, x_2) = 0.$$

If we omit x_2, the simple regression error becomes

$$v = \beta_2 x_2 + u, \qquad \mathbb{E}(v \mid x_1) = \beta_2 \mathbb{E}(x_2 \mid x_1).$$

- If x_2 affects y and its conditional mean varies with x_1, then $\mathbb{E}(v \mid x_1)$ varies with x_1: the simple-regression key assumption fails.
- Multiple regression places x_2 in the model instead of leaving it in the error term.

The direction of omitted-variable bias

Suppose x_2 belongs in the population model but we regress y only on x_1. Write the population linear projection of x_2 on x_1 as

$$x_2 = \delta_0 + \delta_1 x_1 + r, \qquad \delta_1 = \frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)}.$$

Under $\mathbb{E}(u \mid x_1, x_2) = 0$, the simple-regression population slope is

$$\widetilde{\beta}_1 = \beta_1 + \beta_2 \delta_1.$$

The bias has the sign of $\beta_2 \delta_1$: it depends on how x_2 affects y and how x_2 varies with x_1. This result needs finite second moments and $\mathrm{Var}(x_1) > 0$.

With several omitted regressors, their bias contributions can offset one another; this two-variable sign rule is then insufficient.

From the population to a random sample

Let $\{(y_i, x_{i1}, \ldots, x_{ik}) : i = 1, \ldots, n\}$ be a random sample:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i.$$

Stack the n equations as $y = X\beta + u$, where

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}, \qquad \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_k \end{bmatrix}, \qquad u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

$y, u \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times (k+1)}$, and $\beta \in \mathbb{R}^{k+1}$.

Matrix review: operations and inverses

For conformable matrices A, B (inner dimensions match, so AB exists), transposition reverses the order:

$$(AB)' = B'A'.$$

A square matrix $(p \times p)$ is *symmetric* when $A' = A$. Such a matrix A is *regular* (invertible) when a matrix A^{-1} satisfies

$$A^{-1}A = AA^{-1} = I.$$

For regular square matrices of the same size,

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^{-1})' = (A')^{-1}.$$

The explicit formula $A^{-1} = \operatorname{adj}(A)/\det(A)$ applies when $\det(A) \neq 0$; $\operatorname{adj}(A)$ is the transpose of the cofactor matrix. In OLS, $X'Xb = X'y$ is a linear system; software solves the least-squares problem numerically, rather than using cofactors.

Matrix review: quadratic forms and covariance

A symmetric $p \times p$ matrix A is *positive semidefinite* if $a'Aa \geq 0$ for every $a \in \mathbb{R}^p$, including $a = 0$. It is *positive definite* if $a'Aa > 0$ for every nonzero a.

For a random vector $z \in \mathbb{R}^p$, its variance–covariance matrix is

$$\mathrm{Var}(z) = \mathbb{E}\big[(z - \mathbb{E}z)(z - \mathbb{E}z)'\big].$$

Its diagonal entries are variances, its off-diagonal entries are covariances, and it is symmetric and positive semidefinite. For a fixed matrix C,

$$\mathrm{Var}(Cz) = C\,\mathrm{Var}(z)\,C'.$$

In OLS, $a'X'Xa = \|Xa\|^2$. Thus $X'X$ is positive definite and invertible when X has full column rank.

Ordinary least squares

OLS chooses coefficients that minimize the sum of squared residuals:

$$\hat{\beta} = \arg\min_b S(b), \qquad S(b) = (y - Xb)'(y - Xb).$$

Since $(X'X)' = X'X$, the quadratic-term derivative is

$$\nabla_b (b'X'Xb) = \left[X'X + (X'X)'\right] b = 2X'Xb.$$

Thus $\nabla_b S(b) = -2X'y + 2X'Xb$; setting it to zero gives $X'(y - X\hat{\beta}) = 0$.

If X has full column rank,

$$\boxed{\hat{\beta} = (X'X)^{-1}X'y}.$$

The next slide shows how the scalar conditions become this matrix equation.

First-order conditions: from scalar to matrix form

With $\boldsymbol{x}_i = (1, x_{i1}, \ldots, x_{ik})'$ and $\hat{u}_i = y_i - \boldsymbol{x}_i'\hat{\beta}$, differentiate and divide by -2:

$$\sum_i \hat{u}_i = 0, \qquad \sum_i x_{ij}\hat{u}_i = 0 \quad (j = 1, \ldots, k).$$

Stack the $k+1$ conditions into one vector while keeping the sum over i:

$$\sum_{i=1}^{n} \underbrace{\begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{ik} \end{bmatrix}}_{\boldsymbol{x}_i} (y_i - \boldsymbol{x}_i'\hat{\beta}) = 0.$$

The rows of X are $\boldsymbol{x}_i'$, so $\sum_i \boldsymbol{x}_i\hat{u}_i = X'\hat{u}$:

$$X'(y - X\hat{\beta}) = 0 \quad \Longleftrightarrow \quad X'X\hat{\beta} = X'y.$$

These sample conditions mirror $\mathbb{E}(u) = 0$ and $\mathbb{E}(x_j u) = 0$: an example of the method of moments.

Useful OLS equations

Assume X has full column rank. The normal equations give

$$\hat{\beta} = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u.$$

Substitute into the fitted-value and residual definitions:

$$\hat{y} = X\hat{\beta} = \underbrace{X(X'X)^{-1}X'}_{P_X} y,$$

$$\hat{u} = y - \hat{y} = \underbrace{[I_n - X(X'X)^{-1}X']}_{M_X} y = M_X u.$$

P_X projects onto the columns of X; M_X removes that fitted part. The geometric interpretation is developed in the appendix.

OLS fitted values, residuals, and sample properties

Define fitted values and residuals in scalar and matrix notation:

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^{k} \hat{\beta}_j x_{ij}, \qquad \hat{u}_i = y_i - \hat{y}_i, \qquad \hat{y} = X\hat{\beta}, \quad \hat{u} = y - \hat{y}.$$

The $k+1$ first-order conditions imply:

- residuals sum to zero: $\sum_i \hat{u}_i = 0$;
- each regressor has zero sample covariance with the residuals: $n^{-1}\sum_i (x_{ij} - \bar{x}_j)\hat{u}_i = 0$;
- fitted values have zero sample covariance with the residuals: $n^{-1}\sum_i (\hat{y}_i - \bar{y})\hat{u}_i = 0$;
- the point of sample means lies on the fitted hyperplane: $\bar{y} = \hat{\beta}_0 + \sum_{j=1}^{k} \hat{\beta}_j \bar{x}_j$.

These are exact algebraic properties of OLS on any sample.

The meaning of holding other factors fixed

For the fitted equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k,$$

changing x_1 while the other included regressors are *held fixed* gives

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1. \tag{3.18}$$

Multiple regression lets us mimic, with nonexperimental data, a laboratory comparison in which included factors are kept fixed.

Example. In a wage equation with education and experience, $\hat{\beta}_1$ compares predicted wages for workers with the same experience but different levels of education.

When regressors are functionally related

If the model contains income and income squared, they cannot be changed separately. For $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 inc + \hat{\beta}_2 inc^2$,

$$\frac{d\hat{y}}{d\, inc} = \hat{\beta}_1 + 2\hat{\beta}_2 inc.$$

A "partialling out" interpretation of multiple regression

Start from the fitted equation

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_k x_{ik}.$$

One way to express $\hat{\beta}_1$ is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} \hat{r}_{i1} y_i}{\sum_{i=1}^{n} \hat{r}_{i1}^2}.$$

- Regress x_1 on $\boldsymbol{x}_{-1} = (1, x_2, \ldots, x_k)$ and retain the residuals $\hat{r}_{i1}$.
- This provides $\hat{\beta}_1$'s partial-effect interpretation:
  - $\hat{r}_{i1}$ is the part of x_{i1} not linearly explained by $\boldsymbol{x}_{-1}$;
  - $\hat{\beta}_1$ measures the sample relationship between y and x_1 after $\boldsymbol{x}_{-1}$ has been partialled out.
- Multiple regression uses the part of x_1 not linearly explained by the controls.
- For practical purposes, $\hat{\beta}_1$ measures the change in y from a one-unit increase in x_1, holding the other regressors fixed.

Goodness of fit

As in simple regression, OLS decomposes the sample variation in y:

$$\mathrm{SST} = \sum_i (y_i - \bar{y})^2, \quad \mathrm{SSE} = \sum_i (\hat{y}_i - \bar{y})^2, \quad \mathrm{SSR} = \sum_i \hat{u}_i^2,$$

$$\mathrm{SST} = \mathrm{SSE} + \mathrm{SSR}, \qquad R^2 = \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}}.$$

- $100R^2$ is the percentage of sample variation in y explained by the fitted multiple-regression equation.
- Adding a regressor cannot reduce R^2 because OLS can assign it a zero coefficient.
- A high R^2 does not establish correct specification or causality.

Assumptions for unbiasedness

MLR.1 The model is linear in parameters.

MLR.2 The pairs (y_i, x_i) are independent draws from the same population distribution (i.i.d.).

MLR.3 No perfect collinearity: $\text{rank}(X) = k + 1$.

MLR.4 Zero conditional mean: $\mathbb{E}(u \mid x) = 0$. (Together with MLR.2: $\mathbb{E}(u \mid X) = 0$.)

Terminology: under MLR.4, the explanatory variables are *exogenous*; if it fails, at least one is *endogenous*.

Then

$$\mathbb{E}(\hat{\beta} \mid X) = \beta.$$

Unbiasedness is a repeated-sampling statement. It does not say that the estimate from one sample equals the population parameter.

Deriving unbiasedness

Substitute $y = X\beta + u$ into the OLS formula:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'(X\beta + u) \\ &= \beta + (X'X)^{-1}X'u.\end{aligned}$$

Conditional on the observed regressors,

$$\mathbb{E}(\hat{\beta} \mid X) = \beta + (X'X)^{-1}X'\mathbb{E}(u \mid X) = \beta.$$

This decomposition also explains why estimates vary across samples:

$$\underbrace{\hat{\beta} - \beta}_{\text{estimation error}} = (X'X)^{-1}X'u.$$

Even under zero conditional mean, chance can make $X'u$ nonzero in one sample: the estimation error is zero on average, not in every sample. We next study its variance.

Estimating the error variance

Homoskedasticity (MLR.5): $\mathrm{Var}(u \mid x_1, \ldots, x_k) = \sigma^2$.

With random sampling, for the sample:

$$\mathrm{Var}(u \mid X) = \sigma^2 I_n, \qquad \mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1}.$$

For the general OLS problem, $\hat{u}$ has n components and satisfies $k+1$ independent normal equations, leaving $n-k-1$ residual degrees of freedom:

$$\mathbb{E}\left(\sum_i \hat{u}_i^2 \,\middle|\, X \right) = \sigma^2 (n-k-1).$$

Therefore, an unbiased estimator of the error variance is

$$\boxed{\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k-1}}.$$

Estimating the variance of the OLS estimators

Replacing σ^2 by $\hat{\sigma}^2$ gives

$$\boxed{\widehat{\mathrm{Var}}(\hat{\beta} \mid X) = \hat{\sigma}^2 (X'X)^{-1}}.$$

Let $\left[(X'X)^{-1}\right]_{j\ell}$ denote element (j, ℓ). For $j, \ell = 0, \ldots, k$,

$$\widehat{\mathrm{Var}}(\hat{\beta}_j \mid X) = \hat{\sigma}^2 \left[(X'X)^{-1}\right]_{jj}, \qquad \widehat{\mathrm{Cov}}(\hat{\beta}_j, \hat{\beta}_\ell \mid X) = \hat{\sigma}^2 \left[(X'X)^{-1}\right]_{j\ell}.$$

The estimated standard error of $\hat{\beta}_j$ is

$$\mathrm{se}(\hat{\beta}_j) = \hat{\sigma} \sqrt{\left[(X'X)^{-1}\right]_{jj}}.$$

Gauss–Markov theorem

Under linearity, random sampling, full rank, zero conditional mean, and homoskedasticity, OLS is the best linear unbiased estimator (BLUE).

Consider an alternative unbiased linear estimator $\tilde{\beta}$ such that:

- $\tilde{\beta}$ is linear in y: $\tilde{\beta} = Ay$.
- $\mathbb{E}(\tilde{\beta} \mid X) = AX\beta = \beta$, so $AX = I$.

$\hat{\beta}$ is said to be better than $\tilde{\beta}$ if and only if $\mathrm{Var}(\tilde{\beta} \mid X) - \mathrm{Var}(\hat{\beta} \mid X)$ is positive semidefinite.

Equivalently,

$$\forall \lambda \in \mathbb{R}^{k+1}, \qquad \mathrm{Var}(\lambda' \tilde{\beta} \mid X) \geq \mathrm{Var}(\lambda' \hat{\beta} \mid X).$$

Every linear combination $\lambda' \tilde{\beta}$ has variance at least as large as $\lambda' \hat{\beta}$.

Proof of the Gauss–Markov theorem

Let $A_{OLS} = (X'X)^{-1}X'$. Any other linear unbiased estimator can be written as

$$\tilde{\beta} = (A_{OLS} + D)y.$$

If it is unbiased, then $(A_{OLS} + D)X = I$. Since $A_{OLS}X = I$, this implies $DX = 0$.

Under homoskedasticity,

$$\begin{aligned} \operatorname{Var}(\tilde{\beta} \mid X) &= \sigma^2 (A_{OLS} + D)(A_{OLS} + D)' \\ &= \sigma^2 A_{OLS} A_{OLS}' + \sigma^2 DD', \end{aligned}$$

because $A_{OLS}D' = (X'X)^{-1}(DX)' = 0$.

Therefore,

$$\operatorname{Var}(\tilde{\beta} \mid X) - \operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 DD' \succeq 0,$$

since $a'DD'a = \|D'a\|^2 \geq 0$ for every vector a.

Variance of an OLS slope estimator

For $j = 1, \ldots, k$,

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{SST_j(1 - R_j^2)}.$$

- σ^2: more variation left in the error increases the variance. Adding relevant regressors can move factors out of the error and reduce σ^2.
- $SST_j = \sum_i (x_{ij} - \bar{x}_j)^2$: more sample variation in x_j lowers the variance. A larger sample usually increases SST_j.
- R_j^2: the R^2 from regressing x_j on the other regressors. A large R_j^2 means that they explain much of the sample variation in x_j, leaving little independent variation and indicating strong multicollinearity.

What multicollinearity does not do

Perfect collinearity: one column of X is an exact linear combination of the others. Then $X'X$ is singular and the coefficient vector is not unique.

High but imperfect collinearity: the OLS coefficient vector exists, but the partial effects can be imprecise.

- High collinearity does not bias OLS under zero conditional mean.
- It makes it harder to distinguish effects that rely on similar variation.
- Adding an irrelevant regressor does not create bias under zero conditional mean, but it can reduce precision.

Possible responses:

- collect more or more varied data;
- reformulate the question or combine variables when theory permits;
- report the uncertainty rather than deleting controls mechanically.

College GPA: compare two specifications

For 141 students, the outcome is college GPA. `ACT` is an achievement test score.

	Simple OLS	Multiple OLS
Intercept	2.40	1.29
High-school GPA	not included	0.453
ACT score	0.0271	0.0094
Observations	141	141
R^2	not shown	0.176

1. What changes in the comparison represented by the ACT coefficient?
2. Which coefficient would you use to compare students with the same high-school GPA? What does a ten-point ACT difference predict?
3. Why might the simple-regression ACT coefficient be larger?

College GPA: interpreting the change

Holding high-school GPA fixed, ten additional ACT points predict a $10(0.0094) = 0.094$-point difference in college GPA. Holding ACT fixed, a one-point difference in high-school GPA predicts 0.453 GPA points.

The sample omitted-variable identity makes the change in ACT's slope precise:

$$\hat{\beta}_{\mathrm{ACT}}^{\mathrm{simple}} = \hat{\beta}_{\mathrm{ACT}}^{\mathrm{multiple}} + \hat{\beta}_{\mathrm{hsGPA}}^{\mathrm{multiple}} \tilde{\delta}_1,$$

where $\tilde{\delta}_1$ is the slope from regressing `hsGPA` on `ACT`. In this sample,

$$0.0271 - 0.0094 \approx 0.453\,\tilde{\delta}_1, \qquad \tilde{\delta}_1 \approx 0.0391.$$

The identity is exact before rounding the displayed coefficients.

ACT and high-school GPA are positively associated. Excluding high-school GPA therefore raises the ACT slope. Neither equation alone establishes causality.

Summary

- We specify a linear model for the population, with unknown parameters.
- Given a sample, we estimate its parameters by OLS.
- Multiple regression gives partial, ceteris paribus comparisons.
- OLS minimizes the sum of squared residuals and makes each regressor's sample covariance with the residuals equal to zero.
- Zero conditional mean gives unbiasedness; homoskedasticity gives the classical variance formula and Gauss–Markov efficiency.
- Multicollinearity reduces the independent variation used to estimate a partial effect.

Next lecture: standard errors quantify uncertainty. We use them for confidence intervals and hypothesis tests.

Appendix: orthogonal projection

This geometry is in observation space $\mathbb{R}^n$: one coordinate for each observation. For an $n \times p$ design matrix X, its columns span

$$\mathcal{C}(X) = \{Xb : b \in \mathbb{R}^p\} \subseteq \mathbb{R}^n.$$

Each Xb is a possible fitted-outcome vector for the n observations.

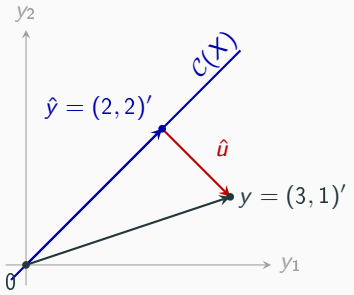
For a subspace $E \subseteq \mathbb{R}^n$, the *orthogonal projection* $P_E y$ is the unique vector in E such that

$$y - P_E y \perp E \quad \text{(that is, } v'(y - P_E y) = 0 \text{ for every } v \in E\text{)}.$$

Equivalently, $P_E y$ is the point of E closest to y in Euclidean distance. The residual is the perpendicular part left after projection.

Appendix: a projection in two dimensions

Simplify to two observations and an intercept only: $X = (1, 1)'$ and $y = (3, 1)'$.



- The axes are the outcomes for observations 1 and 2, not an x-axis and a y-axis.
- $\mathcal{C}(X) = \{(b_0, b_0)' : b_0 \in \mathbb{R}\}$ is the line of possible fitted vectors.
- OLS projects y onto that line: $y = \hat{y} + \hat{u}$, with $\hat{u} = (1, -1)'$ perpendicular to $\mathcal{C}(X)$.

Appendix: OLS is a projection

OLS chooses the closest vector in $\mathcal{C}(X)$:

$$\hat{y} = X\hat{\beta} = P_X y, \qquad P_X = X(X'X)^{-1}X'.$$

The normal equations say $X'\hat{u} = 0$, where $\hat{u} = y - \hat{y}$. Thus $\hat{u}$ is orthogonal to every column of X and

$$y = \underbrace{\hat{y}}_{\in \mathcal{C}(X)} + \underbrace{\hat{u}}_{\in \mathcal{C}(X)^{\perp}}.$$

P_X and $M_X = I_n - P_X$ are symmetric and idempotent:

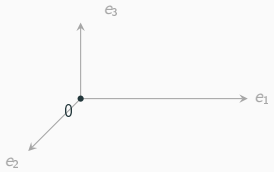
$$P_X' = P_X, \quad P_X^2 = P_X, \qquad M_X' = M_X, \quad M_X^2 = M_X.$$

Projecting a second time does not change either component.

Appendix: projection in observation space

Three observations give three axes. The model has an intercept and one covariate, so $X = [1\ x]$ has two columns.

e_1, e_2, e_3 are the unit directions for observations 1, 2, and 3. Every three-entry vector uses these same axes.

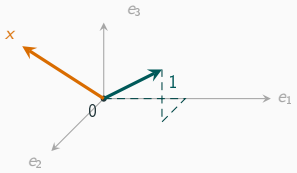


Oblique view of observation space; page angles and lengths are not to scale.

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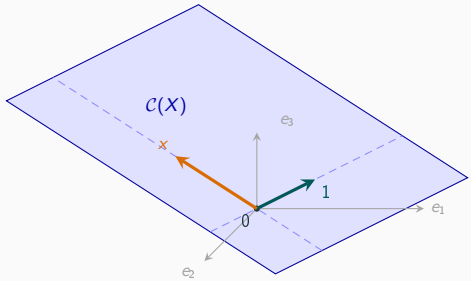
The two arrows are the columns of X: $1 = (1, 1, 1)'$ and $x = (-1, 0, 1)'$. The dashed path adds the three components of 1.



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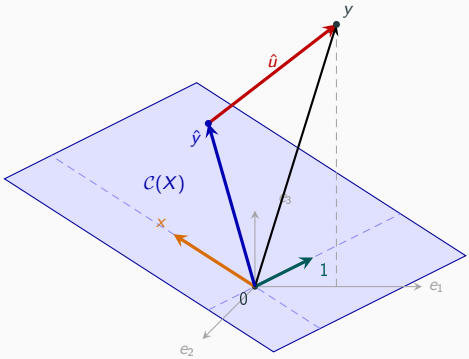


The shaded patch belongs to the unbounded plane $\mathcal{C}(X) = \{b_0 1 + b_1 x\}$. The plane passes through 0.

Oblique view of observation space; page angles and lengths are not to scale.

Appendix: projection in observation space

Three observations give three axes. The model has an intercept and one covariate, so $X = [1\ x]$ has two columns.



Here $y = (1, 0, 5)'$: the dashed path goes one unit along e_1, then five along e_3. The blue $\hat{y} = (0, 2, 4)'$ is its projection. The red $\hat{u} = y - \hat{y}$ is perpendicular to the plane.

Oblique view of observation space; page angles and lengths are not to scale.